ADS-506 Team 1 Final Project - EDA

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```r
library(tidyverse)
library(fpp3)
library(gt)
library(skimr)
library(scales)
```

Dataset

Quick Summary

```r
df <- read_csv("../datasets/total_edify_calls_2024-11-03.csv")
```

```
Rows: 275655 Columns: 8
-- Column specification --------------------------------------------------------
Delimiter: ","
chr  (2): Communication Type, Sub Communication Type
dbl  (4): Wait Time, Time Interacting, Hold Time, Wrap Up Time
dttm (2): Start Time, End Time

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
# remove spaces from column names
colnames(df) <- gsub(" ", "", colnames(df))
head(df)
```

```
# A tibble: 6 x 8
  StartTime           EndTime             CcommunicationType SubCommunicationType
```

```
  <dttm>              <dttm>              <chr>            <chr>
1 2022-03-21 14:13:52 2022-03-21 14:57:26 messaging        text
2 2022-03-21 16:48:33 2022-03-21 17:02:46 messaging        text
3 2022-03-21 16:51:18 2022-03-21 16:57:17 messaging        text
4 2022-03-21 16:57:05 2022-03-21 17:00:37 phone            inbound
5 2022-03-21 17:03:31 2022-03-21 17:04:01 messaging        text
6 2022-03-21 17:09:18 2022-03-21 17:09:49 phone            inbound
# i 4 more variables: WaitTime <dbl>, TimeInteracting <dbl>, HoldTime <dbl>,
#   WrapUpTime <dbl>
```

```r
# quick summary of dataframe
summary(df)
```

```
   StartTime                         EndTime
 Min.   :2022-03-21 14:13:52.00   Min.   :2022-03-21 14:57:26.00
 1st Qu.:2022-11-17 15:41:36.00   1st Qu.:2022-11-17 15:46:31.50
 Median :2023-07-06 22:04:04.00   Median :2023-07-06 22:09:31.00
 Mean   :2023-07-14 15:29:03.66   Mean   :2023-07-14 15:36:04.43
 3rd Qu.:2024-03-12 13:09:51.00   3rd Qu.:2024-03-12 13:13:58.00
 Max.   :2024-10-31 23:57:27.00   Max.   :2024-10-31 23:59:40.00

 CommunicationType  SubCommunicationType    WaitTime       TimeInteracting
 Length:275655      Length:275655        Min.   :      0   Min.   :    0
 Class :character   Class :character     1st Qu.:     10   1st Qu.:   37
 Mode  :character   Mode  :character     Median :     18   Median :   96
                                         Mean   :    119   Mean   :  132
                                         3rd Qu.:     86   3rd Qu.:  177
                                         Max.   :4126179   Max.   :93296
                                                           NA's   :1
    HoldTime          WrapUpTime
 Min.   :   0.000   Min.   :     0.0
 1st Qu.:   0.000   1st Qu.:     5.0
 Median :   0.000   Median :    25.0
 Mean   :   6.317   Mean   :   141.4
 3rd Qu.:   0.000   3rd Qu.:   159.0
 Max.   :1284.000   Max.   :112600.0
 NA's   :1          NA's   :1
```

Columns with missing values

```r
# show columns with missing values
sapply(df, function(x) sum(is.na(x)))
```

```
           StartTime              EndTime    CommunicationType
                   0                    0                    0
SubCommunicationType             WaitTime      TimeInteracting
                   0                    0                    1
            HoldTime           WrapUpTime
                   1                    1
```

```r
# show rows that have missing values
rows_with_na <- df[!complete.cases(df),]
rows_with_na
```

```
# A tibble: 1 x 8
  StartTime           EndTime             CommunicationType SubCommunicationType
  <dttm>              <dttm>              <chr>             <chr>
1 2024-10-05 18:15:15 2024-10-05 18:16:49 phone             inbound
# i 4 more variables: WaitTime <dbl>, TimeInteracting <dbl>, HoldTime <dbl>,
#   WrapUpTime <dbl>
```

```r
# only one row missing values. remove row from dataset
df_clean <- na.omit(df)
df_clean[!complete.cases(df_clean),]
```

```
# A tibble: 0 x 8
# i 8 variables: StartTime <dttm>, EndTime <dttm>, CommunicationType <chr>,
#   SubCommunicationType <chr>, WaitTime <dbl>, TimeInteracting <dbl>,
#   HoldTime <dbl>, WrapUpTime <dbl>
```

```r
# check for rows with NA, Inf, or -Inf
rows_with_non_finite <- df_clean %>%
  filter(
    if_any(c(HoldTime, TimeInteracting, WaitTime, WrapUpTime), ~ !is.finite(.))
  )
rows_with_non_finite
```

```
# A tibble: 0 x 8
# i 8 variables: StartTime <dttm>, EndTime <dttm>, CommunicationType <chr>,
#   SubCommunicationType <chr>, WaitTime <dbl>, TimeInteracting <dbl>,
#   HoldTime <dbl>, WrapUpTime <dbl>
```

Detailed view of data

```r
skim(df_clean)
```

Table 1: Data summary

Name	df_clean
Number of rows	275654
Number of columns	8
Column type frequency:	
character	2
numeric	4
POSIXct	2
Group variables	None

Variable type: character

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
CommunicationType	0	1	5	9	0	3	0
SubCommunicationType	0	1	4	9	0	5	0

Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
WaitTime	0	1	118.69	7975.21	0	10	18	86	4126179	
TimeInteracting	0	1	132.02	303.75	0	37	96	177	93296	
HoldTime	0	1	6.32	34.76	0	0	0	0	1284	
WrapUpTime	0	1	141.40	1034.73	0	5	25	159	112600	

Variable type: POSIXct

skim_variable	n_missing	complete_rate	min	max	median	n_unique
StartTime	0	1	2022-03-21 14:13:52	2024-10-31 23:57:27	2023-07-06 22:03:34	274871
EndTime	0	1	2022-03-21 14:57:26	2024-10-31 23:59:40	2023-07-06 22:08:16	274981

Observation

The dataset has 8 features and 275,655 records.

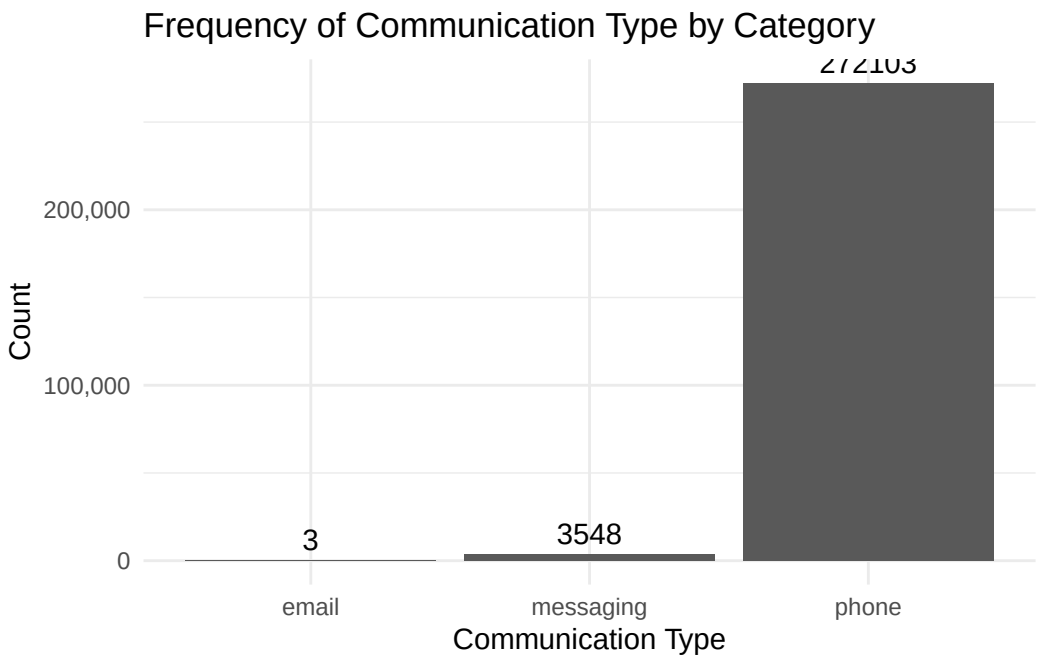
Two of the features are datetime information.

Two features are categorical. Four features are continuous.

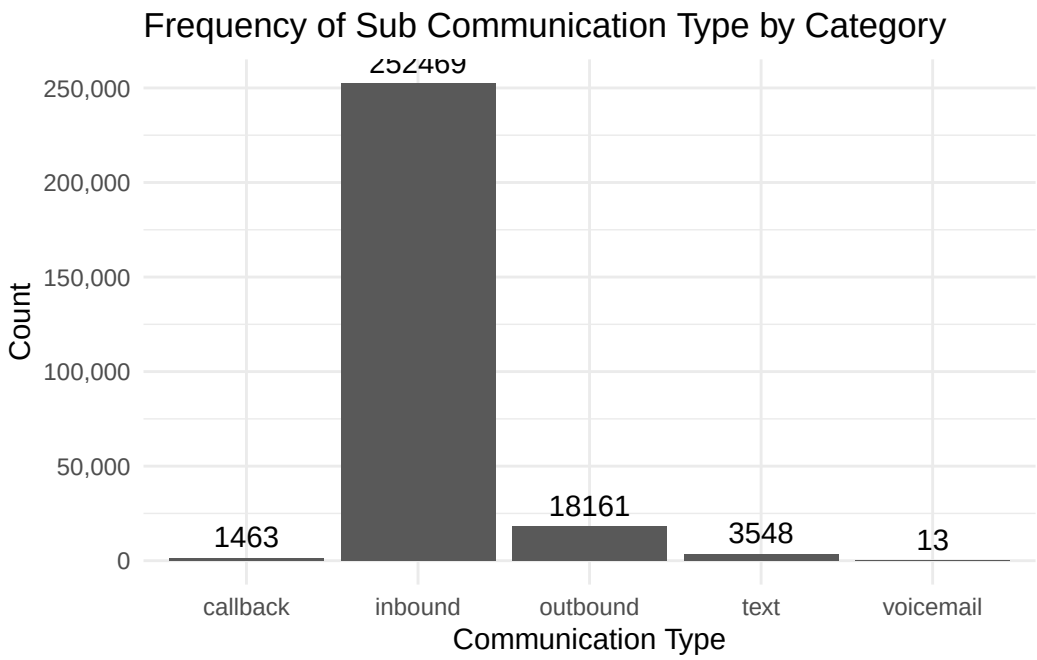
Out of all the records, only one row is missing data.

Categorical Data

```
df_clean %>%
  count(CommunicationType) %>%
  ggplot(aes(x = CommunicationType, y = n)) +
  geom_bar(stat = "identity") +
  geom_text(aes(label = n), vjust = -0.5) +
  labs(
    title = "Frequency of Communication Type by Category",
    x = "Communication Type",
    y = "Count"
  ) +
  scale_y_continuous(labels = comma) +
  theme_minimal()
```




```
df_clean %>%
  count(SubCommunicationType) %>%
  ggplot(aes(x = SubCommunicationType, y = n)) +
  geom_bar(stat = "identity") +
  geom_text(aes(label = n), vjust = -0.5) +
  labs(
    title = "Frequency of Sub Communication Type by Category",
    x = "Communication Type",
    y = "Count"
  ) +
  scale_y_continuous(labels = comma) +
  theme_minimal()
```

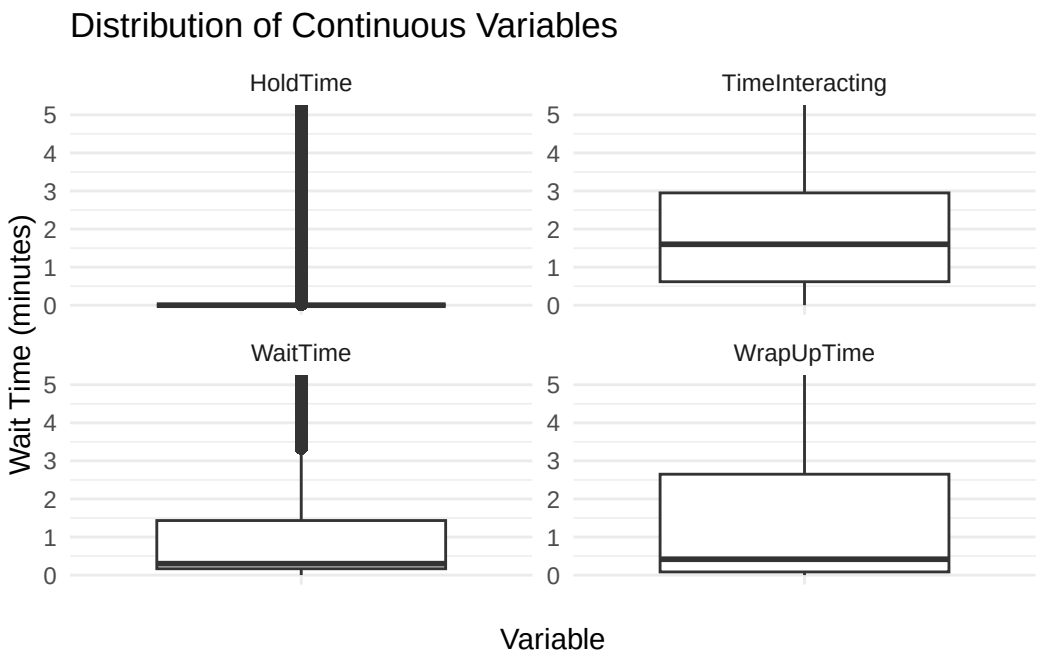


Observations

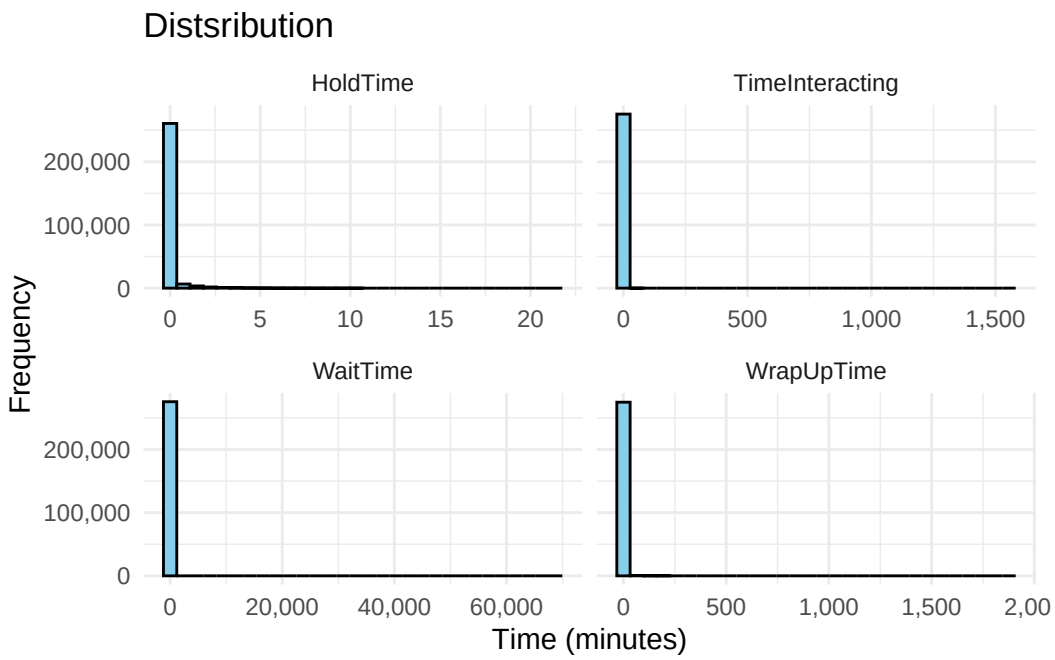
Continuous Variables

```
# reshape data to long format
df_long <- df_clean %>%
  pivot_longer(
    cols = c(WaitTime, TimeInteracting, HoldTime, WrapUpTime),
    names_to = "Variable", values_to = "Value"
  ) %>%
  mutate(Value = Value / 60)

# create box plots
ggplot(df_long, aes(x = "", y = Value)) +
  geom_boxplot() +
  facet_wrap(~ Variable, scales = "free_y") +
  coord_cartesian(ylim = c(0, 5)) +
  labs(
    title = "Distribution of Continuous Variables",
    x = "Variable",
    y = "Wait Time (minutes)"
  ) +
```

```
  scale_y_continuous( labels = comma ) +
  theme_minimal()
```




```
ggplot(df_long, aes(x = Value)) +
  geom_histogram(bins = 30, fill = "skyblue", color = "black") +
  facet_wrap(~ Variable, scales = "free_x") +
  labs(
    title = "Distsribution",
    x = "Time (minutes)",
    y = "Frequency"
  ) +
  scale_y_continuous(labels = comma) +
  scale_x_continuous(labels = comma) +
  theme_minimal()
```




```
continuous_var <- c("WaitTime", "TimeInteracting", "HoldTime", "WrapUpTime")

skim(df_clean[, continuous_var]) %>%
  select(skim_variable, numeric.mean, numeric.sd, numeric.p0, numeric.p25, numeric.p50, numer
  gt() %>%
  fmt_number(
    columns = everything(),
    decimals = 2
  ) %>%
  cols_label(
    skim_variable = "Variable",
    numeric.mean = "Mean",
    numeric.sd = "SD",
    numeric.p0 = "Min",
    numeric.p25 = "25%",
    numeric.p50 = "50%",
    numeric.p75 = "75%",
    numeric.p100 = "Max"
  ) %>%
  tab_header(
    title = "Statistics for Continuous Variables (Minutes)"
  ) %>%
  tab_style(
    style = cell_text(weight = "bold"),
```

Statistics for Continuous Variables (Minutes)

Variable	Mean	SD	Min	25%	50%	75%	Max
WaitTime	118.69	7,975.21	0.00	10.00	18.00	86.00	4,126,179.00
TimeInteracting	132.02	303.75	0.00	37.00	96.00	177.00	93,296.00
HoldTime	6.32	34.76	0.00	0.00	0.00	0.00	1,284.00
WrapUpTime	141.40	1,034.73	0.00	5.00	25.00	159.00	112,600.00

```
    locations = cells_column_labels(everything())
  ) %>%
  cols_align(
    align = "center",
    columns = c(numeric.mean, numeric.sd, numeric.p0, numeric.p25, numeric.p50, numeric.p75,
  )
```

Observations

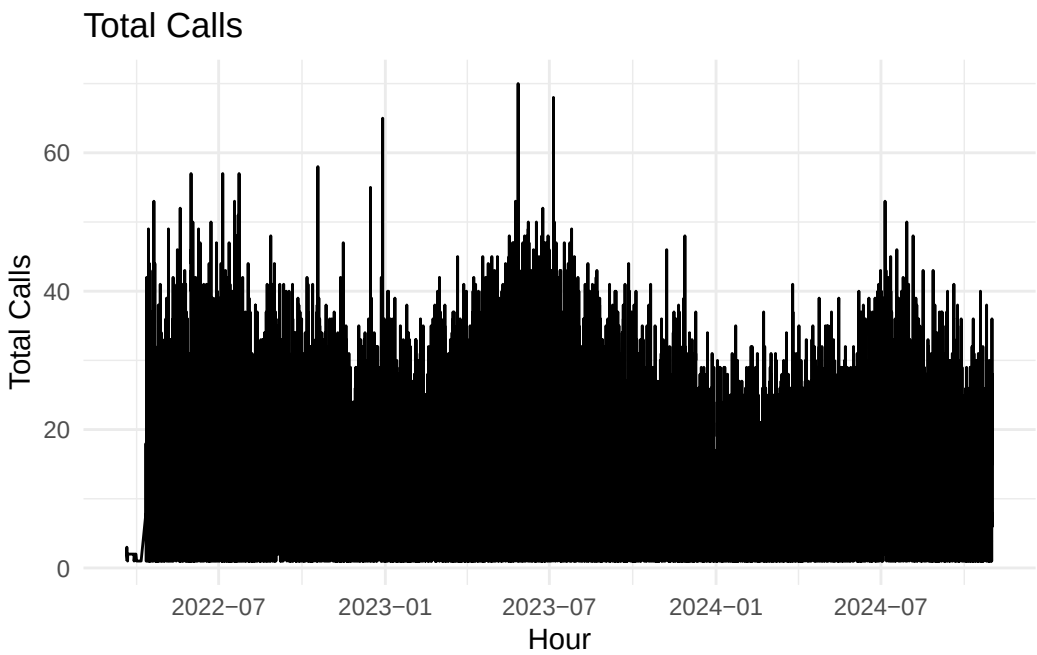
Each of the continuous variables have relatively low means and they also contain extremely high outliers.

Time Series

Hourly

```
# aggregate to hourly
df_hourly <- df_clean %>%
  mutate(hour = floor_date(StartTime, "hour")) %>%
  group_by(hour) %>%
  summarise(total_calls = n())

# convert to tsibble
df_hourly_ts <- df_hourly %>%
  as_tsibble(index = hour)

# plot using autoplot
autoplot(df_hourly_ts, total_calls) +
  labs(
```

```
    title = "Total Calls",
    x = "Hour",
    y = "Total Calls"
  ) +
  theme_minimal()
```

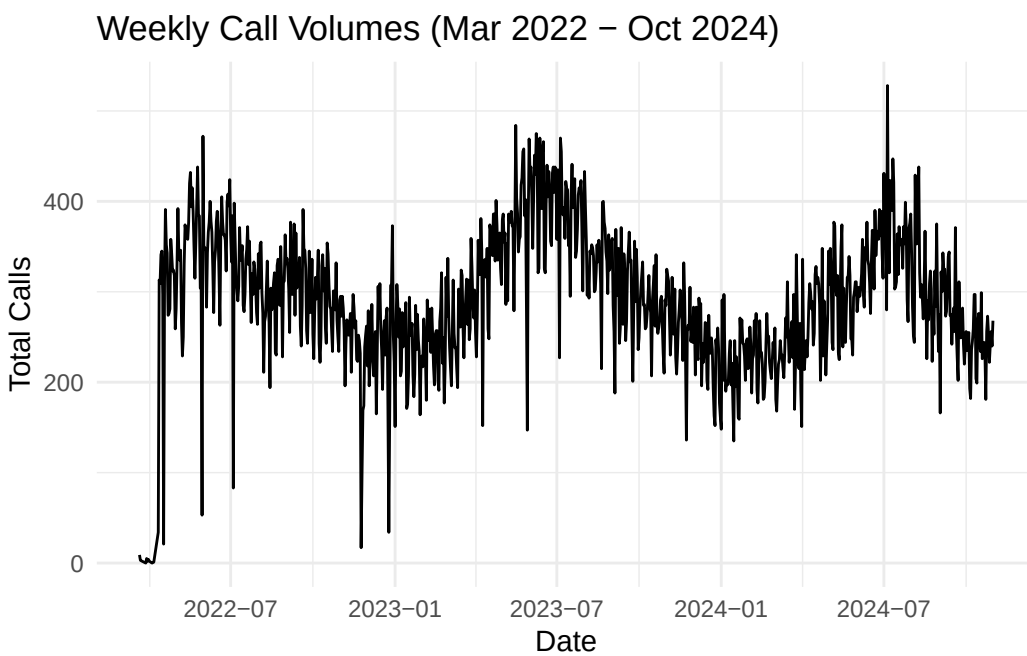


Weekly - Total Call Volume

```
df_weekly <- df_clean %>%
  mutate(week = as.Date(StartTime, "week")) %>%  # Round StartTime to the beginning of the we
  group_by(week) %>%
  summarise(total_calls = n()) %>%  # Count the number of rows (calls) per week
  ungroup() %>%
  # Fill in missing weeks with 0 calls
  complete(week = seq.Date(min(week), max(week), by = "week"), fill = list(total_calls = 0))
```

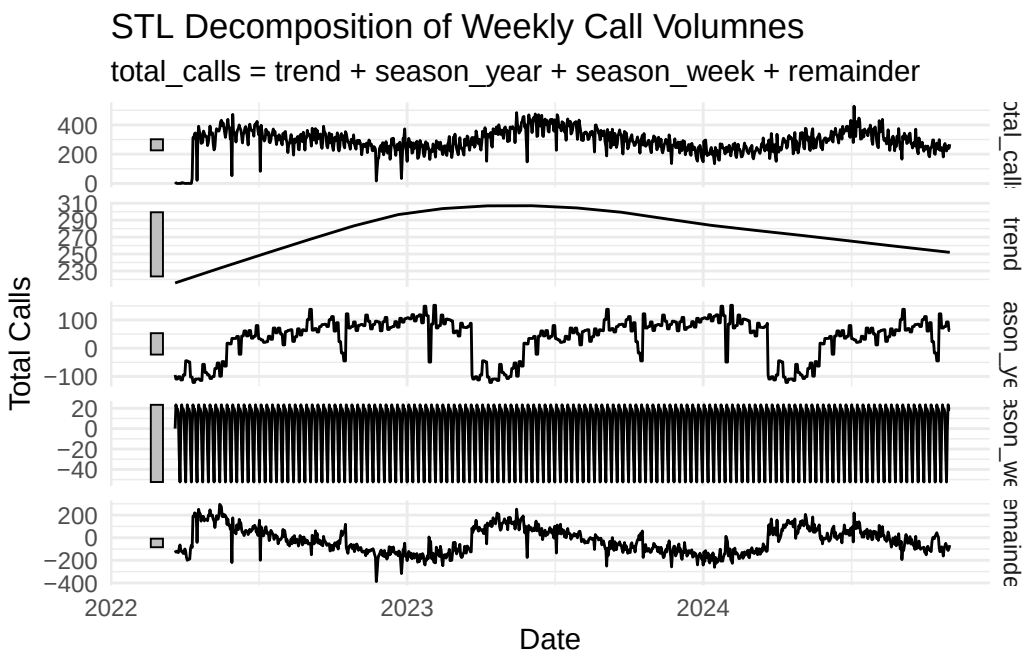
```
Warning: There was 1 warning in `mutate()`.
i In argument: `week = as.Date(StartTime, "week")`.
Caused by warning in `as.POSIXlt.POSIXct()`:
! unknown timezone 'week'
```

```r
df_weekly_ts <- df_weekly %>%
  as_tsibble(index = week)

# plot chart
df_weekly_ts %>%
  autoplot(total_calls) +
  labs(
    title = "Weekly Call Volumes (Mar 2022 - Oct 2024)",
    y = "Total Calls",
    x = "Date"
  ) +
  theme_minimal()
```




```r
# decomp of weekly total call volume
decomp_calls <- df_weekly_ts %>%
  fill_gaps(total_calls = 0) %>%
  model(stl = STL(total_calls ~ season(window = "periodic")))

# extract and view decomp components
components_calls <- decomp_calls %>%
  components()

# plot decomp
```

```r
components_calls %>%
  autoplot() +
  labs(
    title = "STL Decomposition of Weekly Call Volumnes",
    y = "Total Calls",
    x = "Date"
  ) +
  theme_minimal()
```



Observations

Time Series dataset is missing some data that requires filling in. To create these charts, missing weeks were arbitrarily filled in with "0".

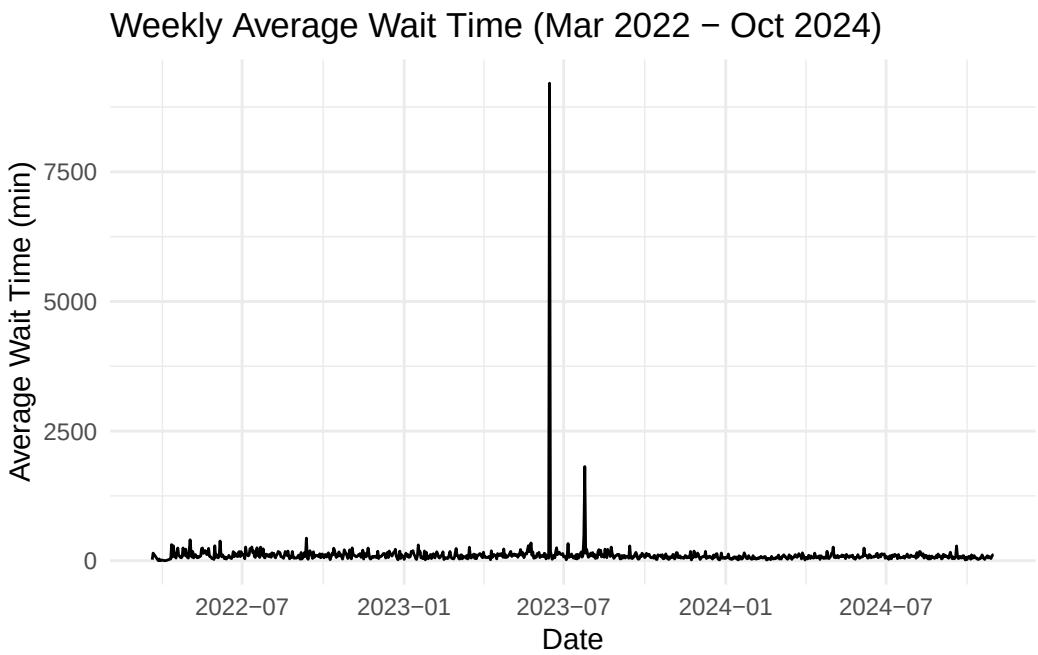
The Remainder chart does not appear to be random. This would need to be explored further to determine if there are any uncaptured season trends, or autocorrelations.

Weekly - Average of WaitTime

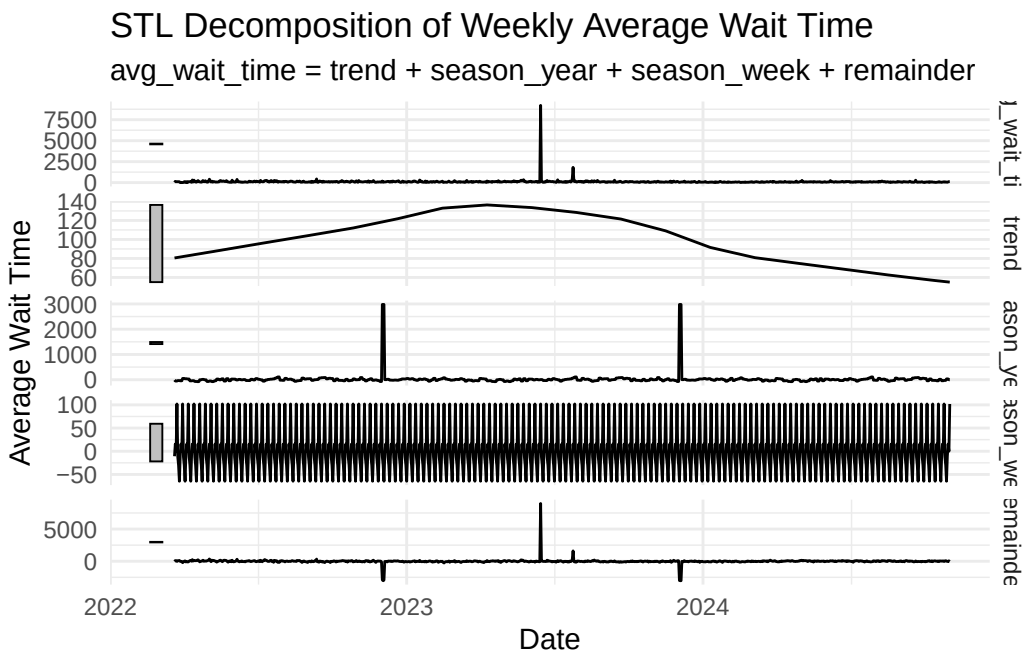
```r
df_weekly_wait <- df_clean %>%
  mutate(week = as.Date(StartTime , "week")) %>%
  group_by(week) %>%
  summarise(avg_wait_time = mean(WaitTime, na.rm = TRUE)) %>%
  ungroup() %>%
  # Fill in missing weeks
  complete(week = seq.Date(min(week), max(week), by = "week"), fill = list(avg_wait_time = 0)
```

```
Warning: There was 1 warning in `mutate()`.
i In argument: `week = as.Date(StartTime, "week")`.
Caused by warning in `as.POSIXlt.POSIXct()`:
! unknown timezone 'week'
```

```r
# convert to tsibble
df_weekly_wait_ts <- df_weekly_wait %>%
  as_tsibble(index = week)

# plot chart
df_weekly_wait_ts %>%
  autoplot(avg_wait_time) +
  labs(
    title = "Weekly Average Wait Time (Mar 2022 - Oct 2024)",
    y = "Average Wait Time (min)",
    x = "Date"
  ) +
  theme_minimal()
```




```
decomp_wait <- df_weekly_wait_ts %>%
  fill_gaps() %>%
  mutate(avg_wait_time = if_else(is.na(avg_wait_time), mean(avg_wait_time, na.rm = TRUE), avg
  model(stl = STL(avg_wait_time ~ season(window = "periodic")))

# extract and view decomp components
components_wait <- decomp_wait %>%
  components()

# plot decomp
components_wait %>%
  autoplot() +
  labs(
    title = "STL Decomposition of Weekly Average Wait Time",
    y = "Average Wait Time",
    x = "Date"
  ) +
  theme_minimal()
```



Observations

Based on average wait time, this does not have any seasonality or cyclic elements. Some weeks were missing data and therefore arbitrarily imputed with the mean wait time.